DIVISION ALGORITHM
&
POWER SET THEOREM

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Well Order Principle: Every non-empty subset of $\mathbb{N}$ has a least element.

Theorem 0.1 (The Division Algorithm). *If a is any integer and b is any positive integer, then there exist unique integers q and r, with $0 \leq r < b$, such that*

$$a = qb + r.$$

Proof. **Existence:** Consider the set

$$S = \{a - nb \mid n \in \mathbb{Z} \text{ and } a - nb \geq 0\}.$$

To see that S is nonempty:

- If $a \geq 0$, then, $a = a - 0 \cdot b \in S$. (taking $n = 0$)
- If $a < 0$, then $a - 2ab = a(1 - 2b) \in S$. [as b is positive integer, so $1 - 2b < 0$ and then $a(1 - 2b) > 0$.]

Thus, in any case, $S \neq \emptyset$. Therefore, by Well-Ordering Principle, S has a least element; say r. This means that, $r \geq 0$ and $r = a - qb$ for some $q \in \mathbb{Z}$.

Now we will show that $r < b$.

To the contrary assume that, $r \geq b$. Then, $r - b \geq 0$, implies $a - qb - b \geq 0$. That is, $a - (q + 1)b \geq 0$.

So, $a - (q + 1)b \in S$ and

$$a - (q + 1)b < a - qb = r,$$

which contradicts the fact that r is the least element of S.

Thus, our assumption is false and therefore, $r < b$.

Thus, we have now found integers q and r such that

$$a = qb + r \quad \text{with} \quad 0 \leq r < b.$$

Uniqueness: Now we will show that q and r are unique.

Suppose, there exists integers q_1, r_1 and q_2, r_2 such that

$$a = q_1 b + r_1 \text{ with } 0 \leq r_1 < b \text{ and}$$
$$a = q_2 b + r_2 \text{ with } 0 \leq r_2 < b.$$

Then, $q_1 b + r_1 = q_2 b + r_2$, which implies

$$(r_1 - r_2) = (q_2 - q_1)b. \tag{0.1}$$

Again, from $0 \leq r_1 < b$ and $0 \leq r_2 < b$ we have, $-b < r_1 - r_2 < b$. Then, $-b < (q_2 - q_1)b < b$, implies $-1 < q_2 - q_1 < 1$. Since $q_2 - q_1$ is an integer, so $q_2 - q_1 = 0$, implies $q_1 = q_2$. Then, from equation (0.1) $r_1 = r_2$. Consequently, q and r are unique. □

Theorem 0.2 (Fundamental Theorem of Arithmetic)**.** *Every integer $n(n > 1)$ can be expressed uniquely as a product of prime numbers.*

For example: $210 = 2 \times 3 \times 5 \times 7$, $100 = 2 \times 2 \times 5 \times 5$.

Theorem 0.3 (Power Set Theorem or Cantor's Theorem)**.** *There exists no onto mapping from a set A to its power set $\mathcal{P}(A)$.*

Proof. We prove the theorem by contradiction.

Suppose that, there exists an onto mapping $f : A \to \mathcal{P}(A)$.

Let, $a \in A$. Then, $f(A) \in \mathcal{P}(A)$. That is, $f(A) \subseteq A$.

[The element a may or may not belong to $f(a)$] $\cdots (*)$

Now, we construct a set $B = \{a \in A : a \notin f(a)\}$. Then, clearly $B \subseteq A$ and so $B \in \mathcal{P}(A)$.

Since, $f : A \to \mathcal{P}(A)$ is onto and $B \in \mathcal{P}(A)$, so there exists $b \in A$ such that such that $f(b) = B$. [by definition of onto mapping].

Since, $b \in A$ and $B \subseteq A$, so b may or may not belong to B.

Now, if $b \in B$ then, $b \in f(b)$ [as $B = f(b)$]. Then from the construction of B, it follows that $b \notin B$, which is a contradiction.

Similarly, if $b \notin B$, then $b \notin f(B)$ [as $b = f(b)$]. Then from the construction of B, it follows that, $b \in B$, which is a contradiction.

So, our assumption is wrong and hence no such $f : A \to \mathcal{P}(A)$ exists such that f is onto. □

Note 0.4. **Understanding of the statement (*) by example:**

Suppose, $A = \{1, 2, 3\}$.

Then, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$. If we define a map $f : A \to \mathcal{P}(A)$ by $f(1) = \{2, 3\}$, $f(2) = \{1, 2\}$, $f(3) = \{2\}$ then

(1) if $a = 2$ then, $a \in f(a)$.
(2) if $a = 1$ then, $a \notin f(a)$.